

H.M. SARWER ALAM

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OBJECTIVE

Computer vision and machine learning researcher interested in research-based higher study. My work focuses on medical image analysis, explainable AI, and federated learning, with a long-term goal of building useful AI systems for healthcare and real-world applications.

EXPERIENCE

- **BNX Network Inc.** 🌐 April 2025 – Present
Software Engineer (Junior) (formerly Trainee Software Engineer) Dhaka, Bangladesh
Responsibilities
 - Built end-to-end machine learning systems, including data processing, model experimentation (Random Forest, XGBoost, SVM, Isolation Forest), training, and deployment using FastAPI.
 - Developed applied ML systems such as network anomaly detection and chatbot applications, and integrated models into Django and C# .NET applications.
- **BRAC University** 🌐 May 2023 - April 2024
Student Tutor (Undergraduate Teaching Assistant), Computer Science and Engineering Dhaka, Bangladesh
Responsibilities
 - Provided 12-hour weekly tutoring on programming, debugging, and problem-solving, and offered personalized guidance to support student learning.**Allocated Courses:** Data Structure (2 semesters), Programming Language-II (1 semester)

EDUCATION

- **BRAC University** 2020 – 2024
Bachelor of Science in Computer Science and Engineering Dhaka, Bangladesh
CGPA: 3.88/4.00 | Vice Chancellor's List (4 semesters), Dean's List (5 semesters)
- **Bir Shreshtha Noor Mohammad Public College** 2019
Higher Secondary School Certificate (HSC) Dhaka, Bangladesh
GPA: 5.00/5.00
- **Bir Shreshtha Noor Mohammad Public College** 2017
Secondary School Certificate (SSC) Dhaka, Bangladesh
GPA: 5.00/5.00

RESEARCH AND PUBLICATION

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [J.1 & T.1] Farhan Faruk, H. M. Sarwer Alam, Rafeed Rahman, Md. Golam Rabiul Alam, Junho Jeong, Md. Kabir Hossain, and Jia Uddin. (2025). **RenSeg: Leveraging Unsupervised Segmentation using Localization and Contour-Guided Quickshift for Renal Calculi and Carcinoma Segmentation and Classification**. *IEEE Journal of Biomedical and Health Informatics (JBHI)*, IEEE Xplore. DOI:10.1109/JBHI.2025.3629580.
Available at: <https://ieeexplore.ieee.org/document/11286221>. **Impact Factor:** 6.8
Contribution: Designed an unsupervised medical image segmentation pipeline ensuring accuracy and reliability in clinical applications.
- [J.2] Tasnim Sakib Apon, Md. Golam Rabiul Alam, Md. Tanzim Reza, Sangita Baidya, Mohammad Fahim Tahmid, Md. Ashraful Alam, Farhan Faruk, H. M. Sarwer Alam, Muhammad Almoyad, Khondokar Fida Hasan, and Mohammad Ali Moni. (2026). **LeukocyteNet: An Explainable Transfer-Transformer Fusion Learning Model for Leukocyte Classification**. *Emerging Science Journal*, 10(3), 1239–1262.
DOI: 10.28991/ESJ-2026-010-03-03. **Available at:** <https://ijournalse.org/index.php/ESJ/article/view/3459>.
Impact Factor: 5.428
Contribution: Co-developed a hybrid CNN-Transformer model with explainable AI for leukocyte classification.
- [C.1 & S] Rakib Hossain Rifat, Farhan Faruk, H. M. Sarwer Alam, Jingjing Yao, and Md. Golam Rabiul Alam. **PLFL: A Semi-supervised Federated Learning Approach Leveraging Pseudo-labeling**. *British Machine Vision Conference (BMVC) 2026*. **Status:** Submitted. **Contribution:** Proposed PLFL, a privacy-preserving federated semi-supervised model achieving 98.34% accuracy on BloodMNIST.

PROJECTS

- **CiteWeave-DuckSummarizer**

Personal Project

Tools: Python, SentenceTransformer, TF-IDF



Developed a web-based text summarization system using hybrid methods combining TF-IDF and embedding-based similarity. Designed a ranking mechanism to extract relevant information and generate concise, cited summaries.

- **Predicting Heart Disease Using Machine Learning**

University Individual Project

Tools: Python, scikit-learn



Developed machine learning models for disease prediction using structured healthcare data. Applied data preprocessing and compared multiple models to analyze performance and improve prediction reliability.

- **Ask My PDF – Local RAG Chatbot**

Personal Project

Tools: Python, FAISS, Transformers



Built a retrieval-augmented generation system for document-based question answering. Designed a pipeline for text extraction, embedding, and retrieval to support interactive and user-driven information access.

SKILLS

- **Programming & Tools:** Python, SQL, C, C#, FastAPI, Django, TensorFlow, PyTorch, scikit-learn, CUDA, Ubuntu, GitHub

- **Core Competencies:** Machine Learning, Model Development and Deployment, Data Processing, Problem Solving, Teamwork, Communication

RESEARCH INTEREST

Interested in computer vision and machine learning for healthcare, with a focus on medical image analysis, explainable AI, and data-driven understanding of human-related signals. Also interested in human-centered and applied AI systems that can support healthcare, rehabilitation, and real-world decision-making. Open to contributing to related research areas such as federated learning, multimodal learning, and trustworthy AI systems.

SCHOLARSHIP AND CERTIFICATION

- **Best Thesis Award - Spring 2024**

Feb 28, 2025

Brac University

Awarded the **Best Thesis Award** for the thesis titled **Leveraging Unsupervised Segmentation for Semi-supervised Renal Calculi and Carcinoma Segmentation and Classification** by the Department of Computer Science and Engineering, School of Data and Sciences, Brac University.

EXTRA CURRICULAR ACTIVITIES

- **Senior Executive - Communications and Marketing**

August, 2021 - August, 2023

BRAC University Computer Club

Led and organized communication and marketing efforts for club events and activities.

- **House Vice Captain - MJ House**

January, 2018 - April, 2019

Bir Shreshtha Noor Mohammad Public College

Oversaw house activities and helped organize events and competitions.

REFERENCES

1. **Md Golam Rabiul Alam, PhD**

Professor of Computer Science & Engineering

BRAC University

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Phone: +8801797347635

Relationship: Thesis Supervisor

2. **Zahin Ahmed**

Graduate Research Assistant (PhD student in CE)

The George Washington University, Washington D.C

Lecturer (on-leave) at Brac University, Dhaka.

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Relationship: Faculty Supervisor for the Data Structure course, in which I served as a Student Tutor